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Structure of a financial cross-correlation matrix under attack

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ABSTRACT

We investigate the structure of a perturbed stock market in terms of correlation matrices. For the purpose of perturbing a stock market, two distinct methods are used, namely local and global perturbation. The former involves replacing a correlation coefficient of the cross-correlation matrix with one calculated from two Gaussian-distributed time series while the latter reconstructs the cross-correlation matrix just after replacing the original return series with Gaussian-distributed time series. Concerning the local case, it is a technical study only and there is no attempt to model reality. The term 'global' means the overall effect of the replacement on other untouched returns. Through statistical analyses such as random matrix theory (RMT), network theory, and the correlation coefficient distributions, we show that the global structure of a stock market is vulnerable to perturbation. However, apart from in the analysis of inverse participation ratios (IPRs), the vulnerability becomes dull under a small-scale perturbation. This means that these analysis tools are inappropriate for monitoring the whole stock market due to the low sensitivity of a stock market to a small-scale perturbation. In contrast, when going down to the structure of business sectors, we confirm that correlation-based business sectors are regrouped in terms of IPRs. This result gives a clue about monitoring the effect of hidden intentions, which are revealed via portfolios taken mostly by large investors.

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1. Introduction

A financial market has been referred to as an example of an adaptive complex system consisting of many interacting components [1,2]. One of the salient features of these complex systems is the spatio-temporal interdependence between components. Accordingly, an abrupt change or variant on a certain component can be spatially dispersed over other components and also affect the others as well as itself temporally. As reported in previous works [1–11], the complicated structure of interdependence among stocks reveals various statistical properties such as scaling structures of returns, a slow decay of returns to Gaussian distributions, a long-ranged property of absolute returns of a market index as an aggregate sum of stock prices, non-universal deviations from the predictions of RMT, etc. Therefore, we can get to the complexity of financial markets by scrutinizing fluctuations of stock quotes and their market index.

Basically, price fluctuations are governed by imbalance between buy and sell orders, which are placed by investors. This imbalance in orders is mainly due to the heterogeneity of investors in expecting the future returns of stocks or a market

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itself. Therefore, most investors are trying to take an optimal portfolio to gain the best profit at the least loss. However, the herding propensity to an aggregate behavior often spoils normal expectations, which are hypothesized by rational expectations theorists. These empirical facts observed in markets underlie the complexity of financial markets. That is, all kinds of interdependence are formed via portfolios, which depend on investors' characteristics, such as preferences, risk aversion, behavioral biases, etc. [12,13].

Portfolios contain investors' characteristics and are also reflected on price fluctuations of stocks. As is well known, all portfolios are composed of a batch of stocks and they often include derivative securities, which are developed to hedge risky assets such as stocks and commodities. Therefore, it is very necessary to understand how the portfolios are constructed and selected. The quantitative analysis of portfolios dates back to the pioneering work of Markowitz in 1952 [13] and, since then, the analysis of the cross-correlation matrix of a financial market has been the critical ingredient in modern portfolio theory [13–16]. In other words, a cross-correlation matrix contains complicated structures of interdependence among investors. The goal of this work is to investigate the vulnerability of the structure of a financial market to perturbations exerted on the market itself. We consider the local perturbation, which is apart from reality. When applying the local perturbation procedure, we simply replace individual elements of the correlation matrix. We point out that such a replacement makes the correlation matrix ill-defined. A consistent procedure would be the replacement of a whole time series r_j which in turn changes all correlation matrix elements C_{ij} with this particular index j . We nevertheless consider the local perturbation for technical reasons. Next, we can examine the effect of a perturbed correlation on a portfolio incorporating several stocks belonging to stock markets geographically apart from the market under consideration. For this purpose, we suggest that making perturbations on the market is performed via replacing several existing components by new time series and reconstructing the correlation matrix, which will be given in more detail in the following section. By utilizing techniques such as RMT analysis, correlation coefficient distributions, and network theory, we obtain significant responses induced from a perturbed correlation matrix constructed from financial time series.

In our work, we investigate the cross-correlation matrix constructed from 492 stocks being exchanged at the Korean stock market in the period 2000–2007. They are all quoted daily. To realize perturbations on the correlation matrix of a stock market, we introduce two methods: replacement of the cross-correlation coefficient of a pair of stocks with one calculated from two Gaussian-distributed time series and reconstruction of a correlation-matrix of the stock market after replacing those returns of two stocks making up a cross-correlation coefficient with two Gaussian-distributed time series. By doing so, we can effectively perturb the structure of a cross-correlation matrix. In Section 2, more details are given about perturbing a correlation matrix. Section 3 give numerical results obtained by applying RMT analysis, correlation coefficient distributions, and network theory to perturbed correlation matrices. Section 4 contains our conclusions.

2. Perturbing a correlation matrix

As briefly mentioned above, perturbation on the correlation matrix of a stock market is performed by replacing correlation coefficients or reconstructing the cross-correlation matrix after replacement. This perturbation, basically, destroys some stylized features observed from financial time series, such as a genuine correlation between two specific stocks, which are supposed to belong to same business group, nonlinear temporal correlations of a singular financial time series, herd behaviors reflected on fat-tails of returns, etc. Therefore, by scrutinizing the structure of a cross-correlation matrix of a financial market, we can get to the qualitative understanding of the relation between a whole stock market and its constituents. In the following, we give a detailed recipe for perturbing the correlation matrix.

Firstly, the local perturbation is made of three steps:

1. Select one of the off-diagonal components of a cross-correlation matrix.
2. Generate two Gaussian-distributed time series and calculate the cross-correlation coefficient from them.
3. Replace the original coefficient with the one obtained from step 2.

In this case, we consider the symmetry of a cross-correlation matrix and thus C_{ij} and C_{ji} are replaced simultaneously. Now, we obtain the locally perturbed cross-correlation matrix. In contrast to the local perturbation, the global perturbation is stronger than that, so the effect of perturbation is more pervasive. Its making follows four steps:

1. Select one of the off-diagonal components of a cross-correlation matrix.
2. Identify two stocks yielding the cross-correlation coefficient of step 1.
3. Generate two Gaussian-distributed time series and replace the original return series of step 2 with newly generated ones.
4. Reconstruct a cross-correlation matrix from the new set of return time series.

By steps 3 and 4, we can induce a global change of a correlation structure if and only if it is very dependent upon a specific stock and that stock is selected from step 1. To this end, we add a rule about performing step 1 in both recipes. Instead of randomly selecting components of a cross-correlation matrix, we take a somewhat systematic approach to selecting those components by sorting correlation coefficients in a descending manner. So, strongly correlated pairs of stocks are top-ranked and weakly correlated ones are bottom-ranked. Now, we can perform step 1 from strongest correlations to weakest ones and vice versa. Actually, this distinction helps in examining the dependence of the correlation-based structure of a stock market on accumulation of stocks and qualitatively proving the existence of leading stocks, which correspond to hubs in terms of network theories. Finally, we can change step 3 of the global perturbation recipe into shuffling original return series to preserve the fat-tailed distribution. As will be shown in the following section, there is nearly no difference.

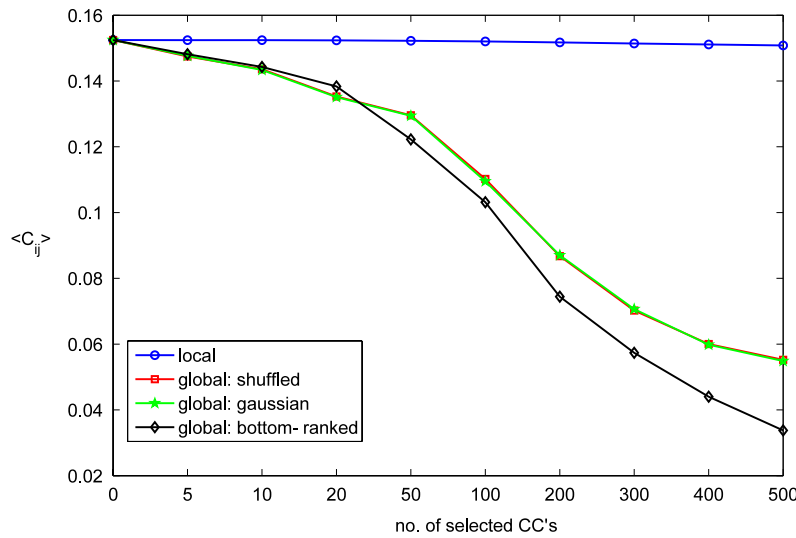


Fig. 1. (color online) Mean values $\langle C_{ij} \rangle$ are presented according to the following perturbation types: local perturbation (circle), global perturbation on strongly correlated pairs of stocks, being shuffled (square) and replaced with an uncorrelated Gaussian (star), and finally global perturbation on weakly correlated pairs of stocks (diamond). For global perturbations, the distinction observed in strongly and weakly correlated pairs is due to the difference of accumulation of nearby stocks. Note: the x-axis represents the number of correlation coefficients selected to be perturbed.

3. Numerical results

To get the analysis started, we first quantify a correlation coefficient between a pair of stocks. All correlation coefficients are calculated from logarithmic returns of financial time series and the logarithmic return, $R_i(t)$, is defined as

$$R_i(t) \equiv \log P_i(t + \Delta t) - \log P_i(t), \quad (1)$$

where $P_i(t)$ denotes the price of stock i and Δt is set to one day. To exclude the effect of a price's magnitude on the correlation coefficient, we use the normalized returns, $r_i(t)$, defined as follows:

$$r_i(t) \equiv \frac{R_i(t) - \langle R_i \rangle}{\sigma_i}, \quad (2)$$

where $\sigma_i \equiv \sqrt{\langle R_i^2 \rangle - \langle R_i \rangle^2}$ is the standard deviation of R_i , and $\langle \cdot \cdot \cdot \rangle$ denotes a time average over the period studied. The correlation coefficient is then computed as

$$C_{ij} \equiv \langle r_i(t)r_j(t) \rangle. \quad (3)$$

The elements C_{ij} are restricted to the range $[-1, 1]$. The following analyses are performed on the cross-correlation matrix constructed as above.

3.1. Distribution of cross-correlation coefficients

Distribution of cross-correlation coefficients is directly related to synchronization of constituents of a complex system, herein stocks of a stock market. In the uncorrelated case, for example, the Wishart matrix gives a Gaussian distribution and shows no explicit co-movements among stocks. However, real financial markets show a clear synchronized movement, i.e., a so-called market-wide effect [3–11]. This empirical observation implies that investors take into consideration information coming from the economy itself when they accordingly adjust their own portfolios. Therefore, when those synchronizing factors are destroyed by putting several stocks into perturbation, distribution of cross-correlation coefficients inevitably responds to those changes. Fig. 1 shows the situation well. The mean value of cross-correlation coefficients represents the degree of synchronization among all stocks. Local perturbations make nearly no effect on the synchronization because only a small fraction of components, estimated at 0.4%, is exposed to being perturbed. On the other hand, global perturbations make a sensitive response to changes and the bottom-ranked pairs of stocks exhibit more sensitivity. This counterintuitive result is explained by considering the accumulation of stocks near strongly and weakly correlated coefficients, as shown in Fig. 2. Strongly correlated pairs of stocks form a strongly connected component in terms of networks, and thus the number of stocks to be perturbed is restricted to a small size, while weakly correlated pairs of stocks cover a large number of stocks.

This analysis shows that the synchronization among stocks in a stock market is mainly due to a high density of stocks near strong correlation coefficients. Fig. 2(b) shows that stocks are packed in a narrow range of strong correlation coefficients. Nevertheless, the degree of desynchronization is similar, as shown in Fig. 1. However, it fails to acquire a high resolution of detection. Figs. 1 and 2(a) suggest that more than a quarter of stocks should be perturbed to detect a half-fall of the mean value of cross-correlation coefficients. This means that a distribution-based means is too poor to monitor the whole stock market affected by a small number of stocks.

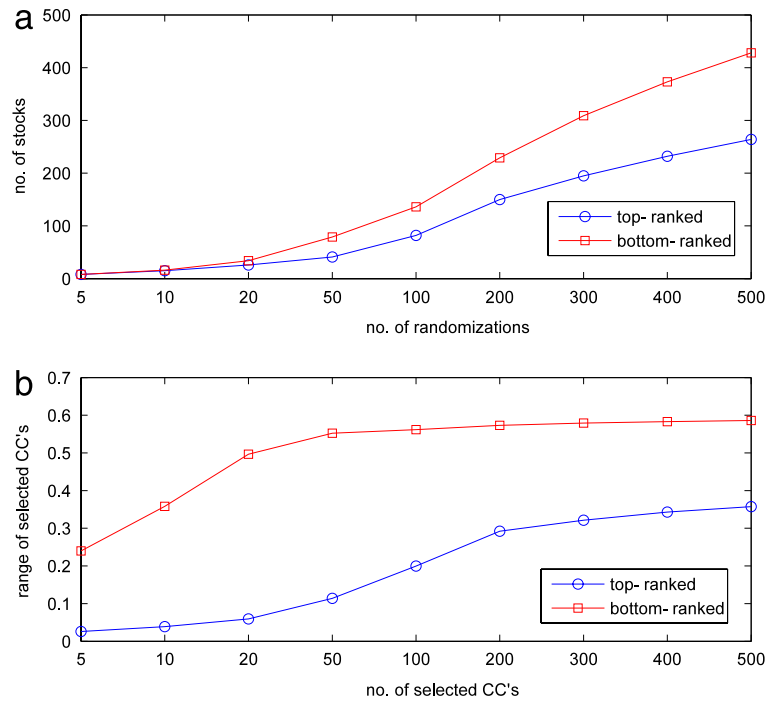


Fig. 2. (color online) (a) Number of stocks corresponding to the selected correlation coefficients and (b) difference between last selected correlation coefficients and first selected ones, respectively, for strongly and weakly correlated pairs of stocks. This plot gives information about the accumulation of stocks near strong and weak correlations, respectively.

3.2. Random matrix approaches to a perturbed correlation matrix

As stated above, understanding the significance and meaning of the cross-correlation coefficients C_{ij} of financial time series is very crucial in allocating one's own wealth. However, the empirical approach to financial time series is always faced with the finite size effect and nonstationarity. RMT was introduced to sort out genuine correlations among stocks from spurious ones, due to reasons mentioned above, through the null hypothesis test. According to previous works performed in recent years [3–11], the financial cross-correlation matrix C is separated into two groups: the part of C conforming to the predictions of RMT and the part deviating from them. Information contained in the deviating part of C is directly related to the structure of the stock market under consideration, the so-called business sectors. As briefly mentioned in Section 3.1, a stock market forms various synchronizing clusters among stocks. These clusters are aggregate of similar or related businesses, for example. Therefore, in this work, it is important to examine the structural changes of the deviating part of C due to perturbation. Before doing so, we confirm that the empirical cross-correlation matrix of the Korean stock market shows properties which are not different from those of previous works [3–6,10,11]. For comparison, we consider a random correlation matrix

$$C_r = \frac{1}{L} A A^T \quad (4)$$

where T denotes the transposition and A is an $N \times L$ matrix containing N time series of L Gaussian-distributed random elements a_{ij} with zero mean and unit variance, that are mutually uncorrelated. In the limit $N \rightarrow \infty$, $L \rightarrow \infty$ with $Q \equiv L/N (> 1)$ fixed, the probability density function $P_{rm}(\lambda)$ is analytically given [17,18] by

$$P_{rm}(\lambda) = \frac{Q}{2\pi\sigma^2} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda} \quad (5)$$

for $\lambda \in [\lambda_-, \lambda_+]$, where λ_- and λ_+ are the minimum and maximum eigenvalues of C_r , respectively, given by

$$\lambda_{\pm} = \sigma^2 \left(1 + \frac{1}{Q} \pm 2\sqrt{\frac{1}{Q}} \right). \quad (6)$$

Fig. 3 shows that the empirical correlation matrix C has two parts, namely the bulk conforming to RMT predictions and the deviating part. For the latter, apart from the largest eigenvalue, genuine correlations among stocks are reconstructed by eigenvalues larger than λ_+ according to

$$C = \sum_{i=1}^{N_r} \lambda_i |\lambda_i\rangle \langle \lambda_i| + \sum_{j=1}^{N_g} \lambda_j |\lambda_j\rangle \langle \lambda_j| + \lambda_{\max} |\lambda_{\max}\rangle \langle \lambda_{\max}|, \quad (7)$$

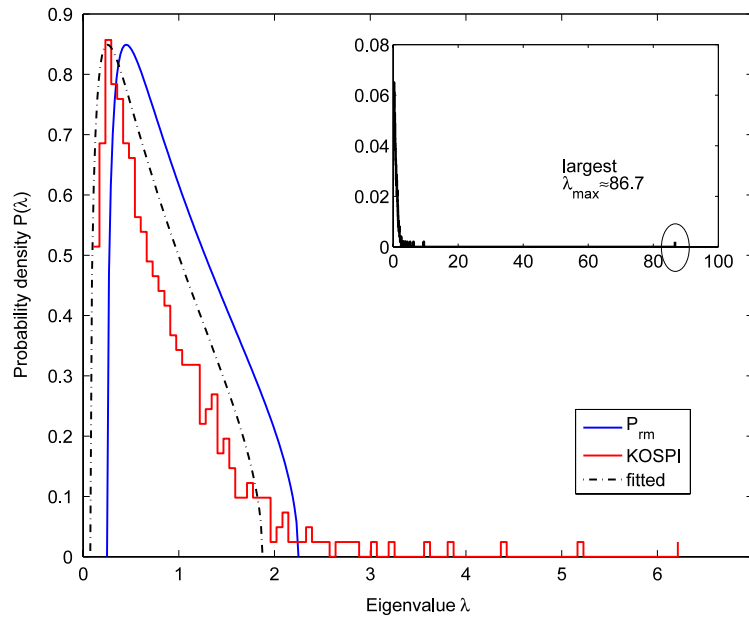


Fig. 3. (color online) The density function of the eigenvalues of a correlation matrix C is shown in comparison with the analytically obtained density function P_m of C_r . The inset shows that the largest eigenvalue is about 40 times larger than the upper bound λ_+ .

where N_r and N_g denote the number of eigenvalues attributed to the random part and the correlations based on the business sector, respectively.

To see the relevance of N_g eigenvalues to business sectors more clearly, we introduce the inverse participation ratio (IPR), often applied in localization theory [19–21]. The IPR of the eigenvector $|\lambda_k\rangle$ is defined as

$$I^k \equiv \sum_{i=1}^N [\langle i|\lambda_k\rangle]^4, \quad (8)$$

where $\langle i|\lambda_k\rangle$, $i = 1, \dots, 492$ are the components of eigenvector $|\lambda_k\rangle$. The meaning of I^k can be illustrated by two limiting cases: (i) all the components are identical, i.e. $I^k = 1/N$, or (ii) only one component has a unit and $I^k = 1$ [20,21]. These examples illustrate how the clustering among stocks can be obtained in terms of the IPR. Through I^k we can obtain the information about how many stocks significantly contribute in financial markets. The inverse participation ratio for the original correlation matrix is presented in Fig. 4. The deviating part shows large I_k 's compared to the bulk while the largest eigenvector has the smallest value, which illustrates the synchronized movement of all stocks due to market-wide effects.

Now, we go back to examining how the deviating part of C responds to perturbations exerted on itself. First, the magnitude $\lambda_{\max}$ of market-wide effects reflected on C is explicitly decreasing, as shown in Fig. 5(a). Also, the number of eigenvalues, each of which is related to a specific business sector, shows the same tendency, shown in Fig. 5(b), with $\langle C_{ij} \rangle$ and $\lambda_{\max}$. Nevertheless, there is still a critical limit to detecting the variation of C due to perturbations by a small number of stocks. In fact, until now, we have focused on the ensemble statistics. However, the disposition of components of an eigenvector $|\lambda_k\rangle$ with a high I_k is stock-dependent. If an arbitrary stock undergoes a specific shock, the cluster this stock belongs to can be rocked or the stock itself can be isolated from its cluster. Fig. 6 shows the variation of the disposition occurring on $|\lambda_{491}\rangle$ for four types of perturbation. It is very natural for the components of an eigenvector of C to be more system-specific compared to other statistics. However, its implication must not be disregarded because this approach is most appropriate for monitoring the whole market.

3.3. Network analysis of a perturbed correlation matrix

Network theory is a very useful tool to explore the skeleton of a system, especially complex systems [22–24]. Without the loss of our aim in this work, we consider the analysis of the vulnerability of a correlation-based network structure to perturbations. Prior to getting started, we must filter out the market-wide effect from C because the synchronization of stocks, due to it, overwhelms the other synchronizations due to the activities of investors based on business sectors. To this end, we adopt the decomposition method forming the basis of widely used economic models, such as multifactor models and the Capital Asset Pricing model [15,16,25]. The decomposition is given as

$$r_i(t) = \alpha_i + \beta_i M(t) + \epsilon_i(t), \quad (9)$$

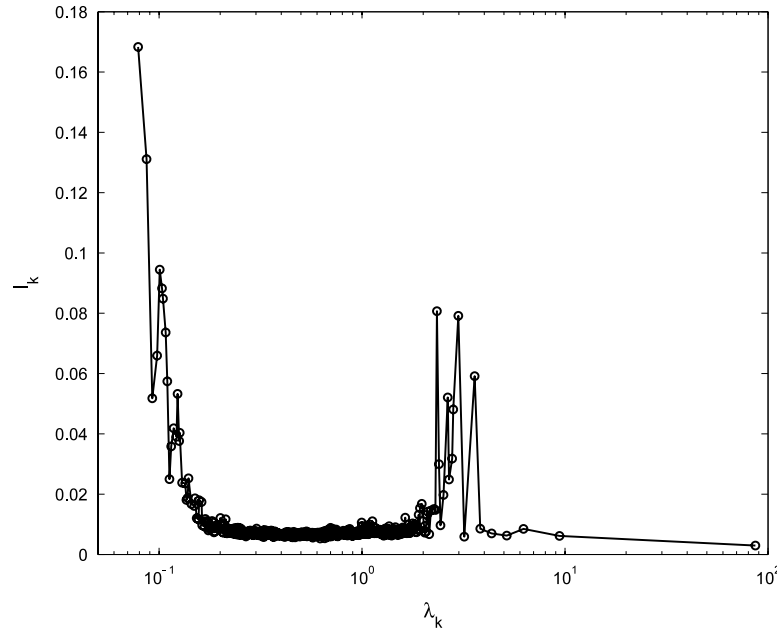


Fig. 4. The inverse participation ratio computed from the correlation matrix of KOSPI is presented in a semilog scale. The bulk part shows the randomness predicted in the random matrix theory while the deviating part shows the evidence of a clustered structure based on business sectors of a stock market. The small value of I^{492} indicates the influences of the market-wide effect on all stocks.

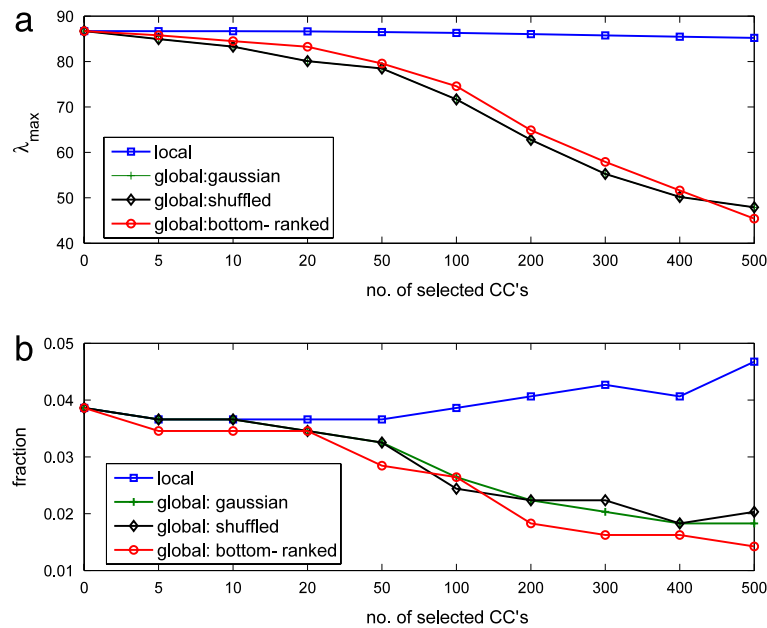


Fig. 5. (color online) The largest eigenvalues with respect to the number of selected correlation coefficients are shown in (a) and the fraction of the eigenvalues deviating from RMT predictions is displayed in (b).

where α_i and β_i are stock-specific constants, and $M(t)$ is a factor giving rise to correlations between stocks. By considering one factor, we approximate $M(t)$ with the projection $r^{492}(t)$ defined as

$$r^{492}(t) \equiv \sum_{j=1}^{492} \langle j | \lambda_{492} \rangle r_j(t) \quad (10)$$

because the components of the largest eigenvector $|\lambda_{\max}\rangle$ quantify market-wide influences on its all stocks. In order to construct a filtered correlation matrix $\mathbf{C}'$, we estimate $\epsilon_i(t)$ by applying the linear square regression to Eq. (8). Next, we determine the threshold value by which the adjacency matrix can be obtained from the filtered correlation matrix $\mathbf{C}'$. The threshold value is set to 0.2, at which the first term and the second term of Eq. (7) are distinguished in terms of distributions of correlation coefficients. Fig. 7 shows the network skeleton versus the strength of perturbation for (a) strongly

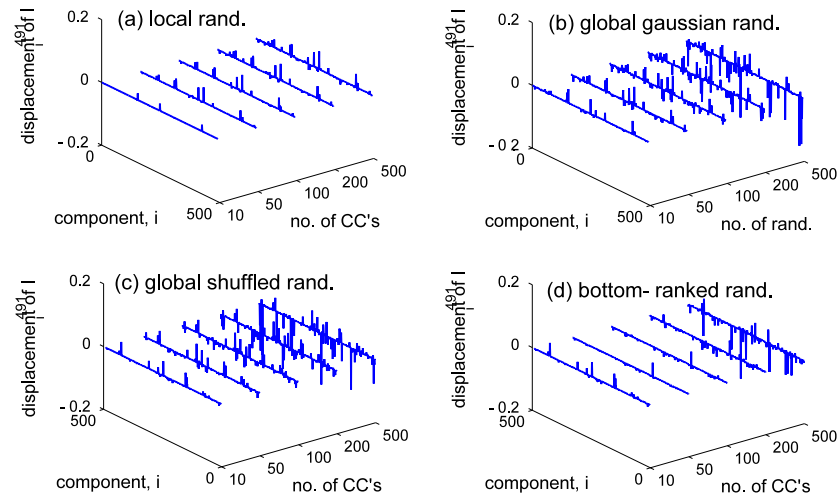


Fig. 6. (color online) The quantities of displacement of each component of $|\lambda_k\rangle$, between $\mathbf{C}$ and its perturbed one, are shown for four types of perturbation: (a) local perturbation, (b) and (c) global perturbation on strongly correlated pairs of stocks, and (d) global perturbation on weakly correlated pairs.

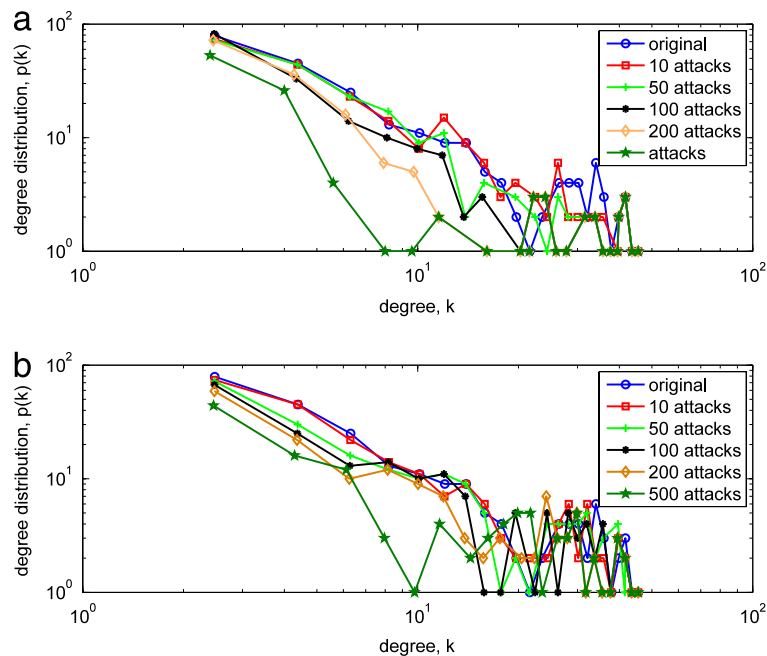


Fig. 7. (color online) The degree distributions obtained from filtered correlation matrices are presented. The power-law behaviors are observed for two types of perturbation: (a) strongly correlated pairs and (b) weakly correlated pairs.

and (b) weakly correlated cases, respectively. Being different from the results of previous analyses, the network skeleton is insensitive to perturbations on the correlation matrix. Fig. 8, also, shows a similar behavior but the evidence of the strong cluster structure among stocks, i.e. perturbing strongly correlated pairs of stocks is more efficient in breaking the links of its network.

4. Conclusions

First, so far, we have examined the vulnerability of the correlation matrix of a stock market, herein KOSPI, through three analysis tools: the distribution of correlation coefficients, random matrix methods, and network theory. Although the results from those analyses show the vulnerable aspects of the correlation matrix to perturbations well, most analysis tools had a limitation in detecting the effect of perturbations on the whole stock market when a tiny fraction of the market is exposed to perturbations. Nevertheless, the stock-specific sensitivity of eigenvectors deviating from the bulk governed by RMT predictions gives a noticeable implication. That is, an IPR analysis can be a useful tool in stock market monitoring if more quantitative means are available. In addition, this wide-scale effect of perturbations on the correlation matrix implies that making complex portfolios incorporating stocks from foreign markets can be more seriously considered based on the results of our work.

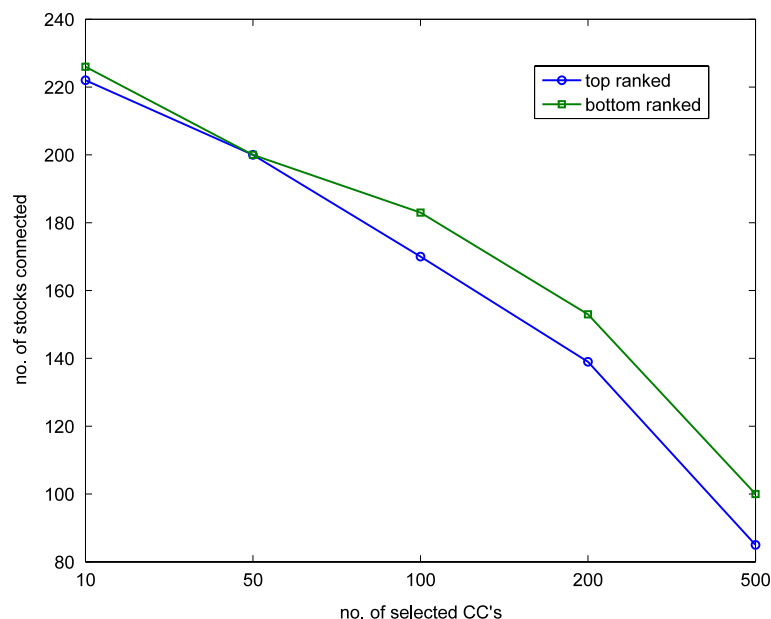


Fig. 8. (color online) The number of connected stocks is plotted. This result shows that pairs of stocks with strong correlations form a strongly connected cluster in a network.

Acknowledgments

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